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Topic 1. Metrics for Performance Evaluation

1. Accuracy

A classification model tested **100 samples**. It correctly classified **85 samples**.

Questions:

1. How many samples were classified incorrectly?
 2. Calculate the accuracy of the model.
 3. Calculate the error rate.
 4. Is the model performing well based on accuracy?
 5. What happens to accuracy if 5 more samples are classified correctly?
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2. Confusion Matrix

A model gives the following results:

- True Positive (TP) = 40
- True Negative (TN) = 35
- False Positive (FP) = 10
- False Negative (FN) = 15

Questions:

1. What is the total number of predictions?
 2. Calculate the accuracy.
 3. Calculate the error rate.
 4. Which value represents incorrectly predicted positive cases?
 5. Which value represents incorrectly predicted negative cases?
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3. Precision and Recall

A disease prediction system produces:

- TP = 80

- FP = 20
- FN = 10
- TN = 90

Questions:

1. Calculate Precision.
 2. Calculate Recall.
 3. Which metric tells us how many predicted positive cases were actually positive?
 4. Which metric tells us how many actual positive cases were correctly identified?
 5. Which metric is more important when missing a positive case is dangerous?
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4. Compare Two Classifiers

Two classifiers are tested on the same dataset.

Metric	Classifier A	Classifier B
Accuracy	90%	85%
Precision	80%	95%
Recall	70%	90%

Questions:

1. Which classifier has higher accuracy?
 2. Which classifier has higher precision?
 3. Which classifier has higher recall?
 4. Which classifier would you choose if detecting as many positive cases as possible is important?
 5. Give one reason why accuracy alone may not be enough.
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5. Analyze Model Performance

A classification model produces the following results:

- TP = 70
- TN = 80
- FP = 20
- FN = 30

Questions:

1. Calculate accuracy.
 2. Calculate precision.
 3. Calculate recall.
 4. Calculate the error rate.
 5. Based on these values, identify one strength and one weakness of the model.
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Topic 2. Prediction Algorithms

1. Simple Linear Prediction

A company's sales are predicted based on advertising expenditure.

Advertising Cost (₹)	Sales
10	20
20	40
30	60
40	80

Assume the relationship is linear.

Questions:

1. Identify the relationship between advertising cost and sales.
 2. Predict sales when advertising cost is ₹50.
 3. What happens to sales when advertising cost increases by ₹10?
 4. Is this an example of classification or prediction?
 5. Why is a linear prediction algorithm suitable here?
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2. Predict Temperature

A simple prediction model is:

Questions:

1. Predict the temperature on Day 5.
 2. Predict the temperature on Day 10.
 3. What is the predicted temperature on Day 15?
 4. How much does the temperature increase for every additional day?
 5. Is this a classification or prediction problem?
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3. House Price Prediction

A prediction algorithm estimates house prices based on area.

Area (sq.ft)	Price (₹ lakhs)
1000	40
1500	60
2000	80

Assume a linear relationship.

Questions:

1. Identify the relationship between area and price.
 2. Predict the price of a 2500 sq.ft house.
 3. What happens to the price when area increases by 500 sq.ft?
 4. Which type of algorithm can be used for this problem?
 5. Is this classification or numerical prediction?
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4. Compare Actual and Predicted Values

A prediction algorithm gives the following results:

Actual Value	Predicted Value
50	48
60	65
70	68
80	85
90	88

Questions:

1. Which prediction has the smallest error?
2. Which prediction has the largest error?
3. Calculate the error for each prediction.
4. Is the model generally predicting close to the actual values?
5. Why is the difference between actual and predicted values important?

5. Choosing a Prediction Algorithm

Consider the following applications:

- A. Predicting tomorrow's temperature.
- B. Predicting the price of a house.
- C. Predicting the number of products sold next month.
- D. Predicting whether an email is spam or not.

Questions:

1. Which applications are **prediction problems**?
2. Which application is a **classification problem**?
3. Which algorithm could be used for predicting house prices?
4. Why is regression suitable for numerical prediction?
5. Give one difference between **classification** and **prediction**.

Answers:

Topic 1. Metrics for Performance Evaluation

1) Accuracy

Given:

- Total samples = 100
- Correctly classified samples = 85

- 1) Incorrect samples = Total samples – Correct samples
= 100 – 85
= 15
15 samples were classified incorrectly.

2) Formula:

$$\begin{aligned}\text{Accuracy} &= (\text{Correctly classified samples} / \text{Total samples}) \times 100 \\ &= (85/100) \times 100 \\ &= 85\%\end{aligned}$$

$$\text{Accuracy} = 85\%$$

3) Formula:

Error Rate = (Incorrectly classified samples/Total samples) × 100

= (15/100) × 100

= 15%

Error rate = 15%

4) The model has an accuracy of **85%**.

Yes, the model is performing **reasonably well**, because it correctly classifies **85 out of 100 samples**. However, **15 samples are still incorrectly classified**, so there is room for improvement.

5) Initially:

- Correct = 85
- Total = 100

If **5 more samples** are correctly classified:

85 + 5 = 90

So the model correctly classifies **90 samples out of 100**.

New Accuracy = (90/100) × 100

= 90%

The accuracy increases from **85% to 90%**.

II) Confusion Matrix

Given Data

A classification model gives the following results:

- True Positive (TP) = 40
- True Negative (TN) = 35
- False Positive (FP) = 10
- False Negative (FN) = 15

1) The total number of predictions is calculated as:

Total Predictions = TP + TN + FP + FN

= 40 + 35 + 10 + 15

= **100**

The total number of predictions is 100.

2) The formula for accuracy is:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \times 100$$

Substituting the given values:

$$\text{Accuracy} = (40 + 35) / (40 + 35 + 10 + 15) \times 100$$

$$\text{Accuracy} = 75 / 100 \times 100$$

$$\text{Accuracy} = 75\%$$

The accuracy of the model is 75%.

3) The formula for error rate is:

$$\text{Error Rate} = (FP + FN) / (TP + TN + FP + FN) \times 100$$

Substituting the values:

$$\text{Error Rate} = (10 + 15) / 100 \times 100$$

$$\text{Error Rate} = 25\%$$

The error rate is 25%.

4) False Positive (FP) represents incorrectly predicted positive cases.

Here:

$$\text{FP} = 10$$

This means that **10 cases were incorrectly predicted as positive when they were actually negative.**

$$\text{False Positive (FP)} = 10.$$

5) False Negative (FN) represents incorrectly predicted negative cases.

Here:

$$\text{FN} = 15$$

This means that **15 cases were incorrectly predicted as negative when they were actually positive.**

$$\text{False Negative (FN)} = 15.$$

III) Precision and Recall

Given Data

A disease prediction system produces the following results:

- True Positive (TP) = 80
- False Positive (FP) = 20
- False Negative (FN) = 10
- True Negative (TN) = 90

1) Precision measures the proportion of predicted positive cases that are actually positive.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \times 100$$

Substituting the given values:

$$\text{Precision} = 80 / (80 + 20) \times 100$$

$$\text{Precision} = 80 / 100 \times 100$$

$$\text{Precision} = 80\%$$

2) Recall measures the proportion of actual positive cases that are correctly identified.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \times 100$$

Substituting the given values:

$$\text{Recall} = 80 / (80 + 10) \times 100$$

$$\text{Recall} = 80 / 90 \times 100$$

$$\text{Recall} = 88.89\%$$

3) Precision tells us how many of the cases predicted as positive were actually positive.

In this case, the precision is **80%**.

4) Recall tells us how many actual positive cases were correctly identified by the model.

In this case, the recall is **88.89%**.

5) When missing a positive case is dangerous, **Recall** is more important.

This is because recall focuses on identifying as many actual positive cases as possible. A high recall means fewer positive cases are missed.

Recall is more important when missing a positive case is dangerous.

IV) Compare Two Classifiers

Given Data:

- Classifier A: Accuracy = 90%, Precision = 80%, Recall = 70%
- Classifier B: Accuracy = 85%, Precision = 95%, Recall = 90%

1) Classifier A has an accuracy of **90%**, while Classifier B has an accuracy of **85%**

2) Classifier A has a precision of **80%**, while Classifier B has a precision of **95%**.

3) Classifier A has a recall of **70%**, while Classifier B has a recall of **90%**.

4) Recall is important when detecting as many positive cases as possible. Classifier B has a recall of **90%**, which is higher than Classifier A's recall of **70%**.

5) Accuracy alone may not be enough because it does not show how well the model identifies positive cases. **Precision and recall** provide additional information about the performance of the classifier.

V) Analyze Model Performance

A classification model produces the following results:

- True Positive (TP) = 70
- True Negative (TN) = 80
- False Positive (FP) = 20
- False Negative (FN) = 30

1) Formula:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \times 100$$

Calculation:

$$\text{Accuracy} = (70 + 80) / (70 + 80 + 20 + 30) \times 100$$

$$\text{Accuracy} = 150 / 200 \times 100$$

$$\text{Accuracy} = 75\%$$

2) Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \times 100$$

Calculation:

$$\text{Precision} = 70 / (70 + 20) \times 100$$

$$\text{Precision} = 70 / 90 \times 100$$

$$\text{Precision} = 77.78\%$$

3) Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \times 100$$

Calculation:

$$\text{Recall} = 70 / (70 + 30) \times 100$$

$$\text{Recall} = 70 / 100 \times 100$$

$$\text{Recall} = 70\%$$

4) Formula:

$$\text{Error Rate} = (\text{FP} + \text{FN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \times 100$$

Calculation:

$$\text{Error Rate} = (20 + 30) / (70 + 80 + 20 + 30) \times 100$$

$$\text{Error Rate} = 50 / 200 \times 100$$

$$\text{Error Rate} = 25\%$$

5) Strength:

The model has an accuracy of **75%** and a precision of **77.78%**, showing that it makes a relatively good number of correct predictions.

Weakness:

The model has a recall of **70%**, meaning that it fails to correctly identify **30% of the actual positive cases**.

- **Strength:** Good accuracy and precision.
- **Weakness:** Recall is only 70%, so some positive cases are missed.

Topic 2: Prediction Algorithms

I) Simple Linear Prediction

Given Data:

Advertising Cost (₹)	Sales
10	20
20	40
30	60
40	80

1) From the given data:

- Advertising cost = ₹10 → Sales = 20
- Advertising cost = ₹20 → Sales = 40
- Advertising cost = ₹30 → Sales = 60
- Advertising cost = ₹40 → Sales = 80

The sales increase proportionally with advertising cost.

The relationship can be represented as:

$$\text{Sales} = 2 \times \text{Advertising Cost}$$

2) Using the relationship:

$$\text{Sales} = 2 \times \text{Advertising Cost}$$

For advertising cost = ₹50:

$$\text{Sales} = 2 \times 50$$

$$\text{Sales} = 100$$

3) From the given data, every ₹10 increase in advertising cost results in a **20-unit increase in sales**.

For example:

$$\text{₹20} \rightarrow \text{Sales} = 40$$

$$\text{₹30} \rightarrow \text{Sales} = 60$$

Therefore:

$$\text{Increase in sales} = 20 \text{ units}$$

4) The output is a **numerical value (sales)**, and the goal is to predict that numerical value based on advertising cost.

5) A linear prediction algorithm is suitable because the given data shows a **straight-line relationship** between advertising cost and sales. Sales increase consistently as advertising cost increases.

II) Predict Temperature

Given Data:

Day	Temperature (°C)
1	25
2	26
3	27
4	28
5	?
6	?
7	?
8	?
9	?
10	?
11	?
12	?
13	?
14	?
15	?

1) Using the pattern:

Temperature = Day + 24

Temperature = $5 + 24$

Temperature = 29°C

2) Temperature = $10 + 24$

Temperature = 34°C

3) Temperature = $15 + 24$

Temperature = 39°C

4) From the given data:

Day 1 = 25°C

Day 2 = 26°C

Day 3 = 27°C

Day 4 = 28°C

The temperature increases by **1°C** for every additional day.

5) The objective is to predict a **numerical temperature value for a particular day.**

III) House Price Prediction

Given Data:

Area (sq.ft)	Price (₹ lakhs)
1000	40
1500	60
2000	80

1) From the given data:

- 1000 sq.ft $\rightarrow$ ₹40 lakhs
- 1500 sq.ft $\rightarrow$ ₹60 lakhs
- 2000 sq.ft $\rightarrow$ ₹80 lakhs

When the area increases by **500 sq.ft**, the price increases by **₹20 lakhs**.

Therefore, there is a **positive linear relationship** between area and price.

2) From the given data, every increase of 500 sq.ft increases the price by ₹20 lakhs.

For 2000 sq.ft:

Price = ₹80 lakhs

For 2500 sq.ft:

Price = ₹80 lakhs + ₹20 lakhs

Price = ₹100 lakhs

3) From the given data:

1000 sq.ft → ₹40 lakhs

1500 sq.ft → ₹60 lakhs

2000 sq.ft → ₹80 lakhs

Therefore, for every **500 sq.ft increase in area**, the price increases by **₹20 lakhs**.

4) Since there is a linear relationship between house area and price, a Linear Regression algorithm can be used.

5) The output is the price of the house, which is a numerical value.

IV) Compare Actual and Predicted Values

Given data:

Actual Value	Predicted Value
50	48
60	65
70	68
80	85
90	88

1) To determine the smallest error, we calculate the absolute difference between the actual and predicted values ($|\text{Actual} - \text{Predicted}|$) for each pair to find the smallest magnitude.

The minimum absolute error is 2, which occurs in three different predictions:

- For actual value 50 and predicted value 48: $|50 - 48| = 2$

- For actual value 70 and predicted value 68: $|70 - 68| = 2$
- For actual value 90 and predicted value 88: $|90 - 88| = 2$

Therefore, the predictions for the actual values of 50, 70, and 90 tie for the smallest error.

2) We look for the maximum absolute difference between the actual and predicted values to find the largest error magnitude.

The maximum absolute error is 5, which occurs in two predictions:

- For actual value 60 and predicted value 65: $|60 - 65| = 5$
- For actual value 80 and predicted value 85: $|80 - 85| = 5$

Therefore, the predictions for the actual values of 60 and 80 tie for the largest error.

3) The error is calculated by subtracting the predicted value from the actual value (Error = Actual – Predicted). The absolute value represents the overall magnitude of that error.

- **For Actual = 50, Predicted = 48:**

$50 - 48 = 2$ (Absolute Error: 2)

- **For Actual = 60, Predicted = 65:**

$60 - 65 = -5$ (Absolute Error: 5)

- **For Actual = 70, Predicted = 68:**

$70 - 68 = 2$ (Absolute Error: 2)

- **For Actual = 80, Predicted = 85:**

$80 - 85 = -5$ (Absolute Error: 5)

- **For Actual = 90, Predicted = 88:**

$90 - 88 = 2$ (Absolute Error: 2)

4) Yes, the model is predicting very close to the actual values. The absolute errors are quite small (either 2 or 5) relative to the overall scale of the data, which ranges from 50 to 90. For instance, an error of 2 on a target value of 50 represents a deviation of only 4%, and an error of 5 on a target value of 60 represents a deviation of about 8.3%. Because all the predictions fall within a narrow margin of the actual figures, the model demonstrates a strong fit and high overall accuracy.

5) The difference between the actual and predicted values, commonly referred to as the residual or error, is essential because it directly quantifies the accuracy and reliability of the prediction model. By analyzing these differences, we can measure how well the algorithm performs, identify if it has a systematic bias (such as consistently overestimating or underestimating), and determine if the model needs mathematical adjustments to improve its future predictions. Without measuring this difference, there is no objective way to trust or evaluate the algorithm.

V) Choosing a Prediction Algorithm

Given:

- A.** Predicting tomorrow's temperature.
- B.** Predicting the price of a house.
- C.** Predicting the number of products sold next month.
- D.** Predicting whether an email is spam or not.

1) Applications **A** (Predicting tomorrow's temperature), **B** (Predicting the price of a house), and **C** (Predicting the number of products sold next month) are all prediction (or regression) problems. In each of these cases, the goal is to forecast a continuous, precise numerical value—such as a specific temperature in degrees, an exact currency amount, or a measurable quantity of inventory.

2) Application **D** (Predicting whether an email is spam or not) is a classification problem. Instead of calculating a continuous number, the algorithm must evaluate the input and assign it to one of two predefined, discrete categories or labels: "Spam" or "Not Spam".

3) A **Linear Regression** or **Multiple Linear Regression** algorithm is commonly used for predicting house prices. If you are predicting the price based on a single factor (like square footage), simple linear regression is appropriate. If you are using multiple factors simultaneously (such as square footage, number of bedrooms, and age of the house), multiple linear regression will model the relationship to output an estimated numerical price.

4) Regression is mathematically designed specifically to find and model the continuous relationship between independent variables (inputs) and a dependent variable (the output). Because its underlying equations map inputs to an infinite scale of numerical values, it can output precise quantities, measurements, or amounts (like 72.5 degrees or ₹50,500) rather than forcing the output into rigid, predefined categories.

5) The fundamental difference lies in the type of output they generate:

- **Prediction (Regression)** outputs a **continuous numerical value**. The result is a precise measurement or quantity along a scale, such as forecasting a sales figure of ₹10,450.
- **Classification** outputs a **discrete label or category**. The result separates data into predefined groups, such as categorizing an image as a "Cat" or a "Dog," with no in-between numerical scale.